

SCI-ALIGN v0.1
Scientific Rationale for Sample Identity, Timing Alignment, and Scan Support
Stage B Targeted Revision r0.3 — Author Draft

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Abstract

SCI-ALIGN establishes one observation-scoped relation between native detector, telescope, and admitted auxiliary occurrences and a stable detector-reference grid. Its scientific job is narrow but foundational: preserve what happened, when it happened under an admitted clock relation, how each target slot was formed, and what support and uncertainty are actually available. It acts on the paired raw KID coordinates x/r only after Tune/readout and before AST, RTC, calibration, correlated-component treatment, validation policy, or map deposition. This narrative explains the scientific logic; the companion Engineering Conformance Specification retains the complete notation, twenty canonical equations, fifty-five requirements, and twenty-six predictions.

Authority status. This is a Stage B targeted author draft, not scientific-owner approval. It does not assess implementation conformity, observational validation, freeze, operational readiness, or production use.

Reading guide. Sections 1–9 follow the scientific questions in the order a reducer encounters them: identity and time, readout ordering, field operators, telescope-state ownership, missing support, interval identity, downstream transfer, and finally response and open decisions.

Shared identity rule. s is the stable ALIGN detector-reference slot and (o, s) is its stable observation-scoped identity. j is a local storage row only. n is reserved for a stable RTC output-sample identity. No downstream contract reconstructs s from j .

1 What ALIGN scientifically establishes

ALIGN creates a checked, immutable account of temporal relationship. It does not make sky coordinates or maps and does not decide whether a later estimator may use a sample. It establishes the detector-reference interface $i_{\text{ref}} = D$, its nominal grid, stable slot identities (o, s) , exact native source relations, field-specific mappings, and local origin, validity, support, exposure, response, uncertainty-availability, and provenance facts.

The central output is therefore a relation, not merely an array. A consumer must be able to distinguish a native occurrence from a nominal slot, an exact copy from interpolation, a numerical invalidity from a missing occurrence, and local ALIGN validity from AST validity or named-use eligibility. Array shape and row order cannot carry those meanings by themselves.

2 Native event time versus nominal detector-reference slot

Each producer owns the event and epoch meaning of its native coordinate. ALIGN may place an admitted interface on detector-reference time only through the checked relation

$$t_{ik}^{\text{ref}} = (E_i - E_{i_{\text{ref}}})_{\text{sec}} + u_{ik} + \delta_{i \rightarrow \text{ref}}, \quad \delta_{\text{ref} \rightarrow \text{ref}} = 0.$$

The offset is observation-constant in v0.1, producer-authorized, expressed in seconds, and applied exactly once. The corrected event time remains distinct from nominal slot time t_s , assignment residual, native integration support, nominal cell support, and acquired exposure.

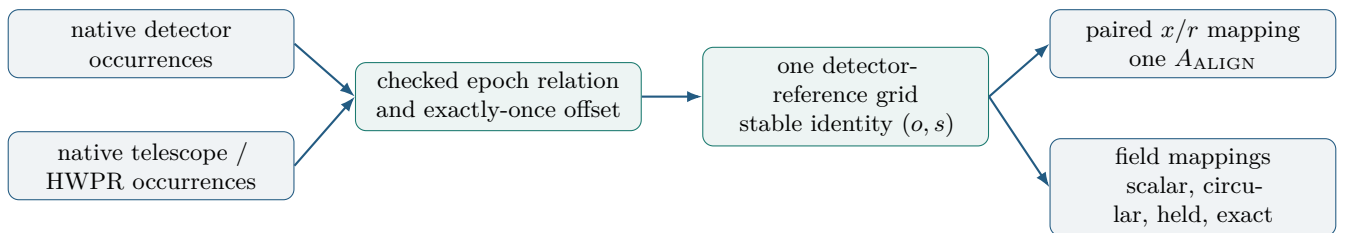


Figure 1: Native occurrences enter one checked time relation before assignment to a stable detector-reference grid. The paired detector coordinates and each registered field then retain their distinct mapping semantics.

3 Tune/readout before paired x/r alignment

The ordinary detector-signal chain is

$$(I, Q)^{\text{acq}} \longrightarrow \text{Tune/readout} \longrightarrow (x, r)^{\text{acq}} \longrightarrow \text{ALIGN} \longrightarrow (x, r)^A.$$

Here I, Q are acquired microwave quadratures, while x, r are the paired physical KID readout coordinates. The symbols x and r are reserved for those coordinates; the detector reference is instead $i_{\text{ref}} = D$. Tune/readout owns the conversion, its units, normalization, and mapping revision. ALIGN neither aligns I, Q first nor assumes that Tune conversion commutes with resampling or synthesis.

One realized temporal operator acts separately on both readout coordinates:

$$x^A = A_{\text{ALIGN}} x^{\text{acq}}, \quad r^A = A_{\text{ALIGN}} r^{\text{acq}}.$$

The outputs share native occurrences, weights, residuals, stable slots, source-relation identity, and origin/synthesis state. Numerical validity and payload availability remain coordinate-specific. One coordinate never fills or rescues the other, and no mapping crosses a Tune or readout-revision boundary without separate authority for that operation, response, and uncertainty.

This distinction also protects astrometry. AST can sometimes construct a coordinate from a valid occurrence/time relation, geometry, and observing state even when one or both numerical x/r payloads are invalid or not requested. Signal-coordinate validity is transferred, but it is not silently promoted into AST coordinate validity or downstream eligibility.

Nonpolarimetric scope. ALIGN v0.1 admits only the ordinary nonpolarimetric detector-signal path. The x/r coordinates remain raw KID readout coordinates. Optional HWPR timing/alignment facts do not authorize demodulation, polarization calibration, or Stokes reconstruction.

4 Field-specific mapping: scalar, circular, held, and exact-only

ALIGN does not apply one generic interpolation rule to every telescope or auxiliary field. Before mapping, each field needs a versioned registry entry that fixes identity, unit, source frame, topology, validity and missing-data meaning, maximum support span, uncertainty availability, and operator.

Field class	Permitted mapping	Scientific guardrail
Continuous scalar	Exact copy at coincidence or bounded adjacent linear interpolation within one unbroken valid run.	No extrapolation and no skipping across invalid rows; unit and frame do not change.
Circular angle	Exact copy or shortest-arc interpolation using $d_P(a, b) \in [-P/2, P/2)$.	Exactly antipodal inputs are unavailable unless an explicit unwrap authority resolves the path.
Declared step state	Hold only inside a producer-authoritative half-open validity interval.	A held value at a non-native time is synthesized and retains held provenance.
Exact-only	Exact coincidence only.	Between native occurrences the field is unavailable rather than guessed.

For scalar interpolation at target slot s , the motivating relation is

$$y_s = (1 - \lambda_s)v_a + \lambda_s v_b, \quad \lambda_s = \frac{t_s - t_a}{t_b - t_a}.$$

Circular interpolation uses the same fraction but applies it to the declared shortest signed difference and then wraps into the field’s declared period. The half-open interval is a representation convention, not permission to choose a path for an exactly antipodal pair.

Timestamp, identity, category, mode, flag, counter, and hardware-state fields cannot inherit scalar treatment from their numerical representation. A missing registry entry blocks only the dependent field or claim and cannot be recovered from a familiar name, shape, or numerical range.

5 Observing-state fields versus pointing-correction records

Two telescope families serve different scientific purposes and must remain separate. The detector-reference occurrence time t_s is the exact grid/time identity, not another scalar observing-state field subject to interpolation. The observing-state family describes the science occurrence at that exact time: aligned boresight, current elevation, required azimuth, and any additional registered values needed by AST geometry or field rotation. ALIGN preserves the producer’s unit, frame, topology, support, and meaning. A producer telescope timestamp may remain native support metadata, but it is not a competing current-sample time.

The pointing-correction family is not another name for current observing state. A pointing-support producer selects an exact correction record and its native support and supplies record, parent, selection, vector-meaning, sign, basis, unit, time/support, covariance-availability, and producer identities. ALIGN contributes only the admitted relation of that selected record to detector-reference time. AST may interpolate a correction only within the selected native support and never extrapolates.

In particular, current science-sample elevation is not inferred from a bracketing correction record. Field rotation uses the aligned occurrence-local observing state. Correction composition, geometry, continuous sky/map coordinates, astrometric uncertainty, and TAN/WCS projection remain AST responsibilities. Physical center selection remains with the named target or product authority.

Optional HWPR fields are a third typed family. Their timing and alignment can be represented without trimming the ordinary nonpolarimetric detector support. Until a future polarimetry authority exists, these facts do not authorize demodulation or Stokes reconstruction.

6 Original values, interpolation, gaps, continuity surrogates, and exposure

A nominal grid cell is not evidence that a detector acquired anything. ALIGN keeps distinct: a native original occurrence, an interpolated field value, an acquisition timing gap, an original occurrence with invalid payload, a processing guard, an unavailable optional stream, and an explicitly authorized detector-signal continuity surrogate.

Ordinary field interpolation follows the registered operator. It does not authorize detector-signal synthesis. A continuity surrogate would need a separate signal-domain authority specifying count, duration, fraction, edge, causal support, response, uncertainty, and guard rules. That authority is unresolved in v0.1, so the conditional gap equation in the engineering view is not enabled. If ever authorized, a surrogate keeps exact endpoints and weights, synthesized origin, and zero added acquired exposure; it does not gain an independent hit, degree of freedom, or significance count.

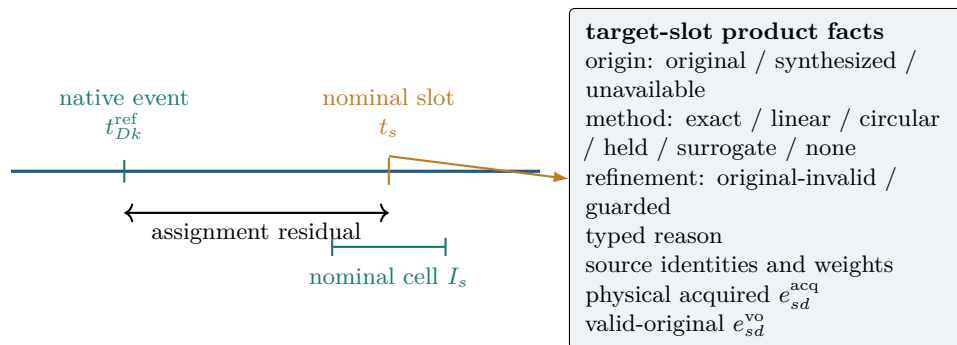


Figure 2: One stable target slot carries distinct timing and product-state facts. Physical acquired exposure records attributable original integration even when its payload is invalid; valid-original exposure is the usable local subset. Synthesized, missing, and unoccupied support add zero acquired exposure, and later guards or use policy do not rewrite acquisition.

7 Physical scans versus processing and science windows

Five interval identities remain separate even when their numerical bounds happen to coincide:

1. producer-authoritative physical scan or state interval;
2. processing chunk;
3. science window;
4. context window; and
5. output selection.

Each is stable, observation-scoped, and half-open. Subsetting or repartitioning does not renumber physical scans or stable ALIGN slots. Short, partial, empty, guarded, and unusable states retain identity rather than being padded away.

A requested fixed-duration partition may define processing windows after its duration is resolved to a sample count. It cannot replace physical scan authority or reinterpret a producer state such as `Hold`. The exact transition and validity semantics for physical scan segmentation remain an owner/producer question.

Within a materialized window, local row j may be derived from stable slot s through the explicit selection relation. That local row is convenient storage, not external identity. The stable cross-package key remains (o, s) , while n remains reserved for RTC output samples. This rule prevents a changed slice boundary or array layout from silently changing scientific identity.

Physical acquired exposure also follows stable source identity rather than window membership. Valid-original, guarded, retained, and use-qualified exposure remain separate later facts. Overlapping membership is rejected unless a successor explicitly defines it and prevents double counting.

8 What AST and RTC receive

AST receives one immutable bundle under profile `SCI-ALIGN_TO_SCI-ASTv0.1/r0.1`. The package manifest binds the final boundary bytes; the boundary body contains no self-hash. The bundle is not a menu of substitutable records. Every required item is present with exact identity and semantics or typed unavailable with reason.

The transfer includes the reference grid and stable (o, s) identity; native and assigned timing; offset authority; exact or generatively reconstructible source relations; the common paired x/r mapping and coordinate-specific validity; observing-state and pointing-correction families; origin, support, physical acquired and valid-original exposure, intervals, mapping validity, response, covariance availability, and complete requested-to-realized provenance. Each materialized view carries its explicit s -to- j relation. AST does not reconstruct clocks, repeat ALIGN policy, extrapolate pointing support, infer missing semantics, or treat a nominal slot or synthesized value as exposure.

RTC receives stable ALIGN identities and relation facts before owning its own temporal conditioning and stable output-sample identity n . CAL owns calibration and atmosphere operations; PTC owns correlated-component estimation; VAL evaluates predicates registered by named-use owners; and MAP owns sample-to-pixel deposition, estimator-specific projection G_π , weights, contribution, and map support. Every consumer preserves ALIGN origin, source support, response/covariance availability, and zero exposure for synthesized detector values.

Compatibility rule. Similar shape or field names do not establish compatibility. A successor must name the profile revision it supersedes and map every changed, removed, or newly required semantic item.

9 Response, covariance, validation, and unresolved owner questions

ALIGN’s response is conditional on the realized mapping. Even a linear interpolator that preserves DC generally changes amplitude and phase away from zero frequency. Gap response, if ever authorized, is row-specific and nonstationary. ALIGN therefore exports an exact conditional response for the admitted tier or a typed reason it is unavailable; it does not claim an end-to-end downstream transfer function.

When a full admitted joint covariance is supplied, an affine realized mapping uses $C_{y|A} = AC_{v|A}A^T$, preserving known cross-time, cross-detector, and cross-stream terms. Timing, interpolation-model, mapping, and selection uncertainties are reported when admitted or typed unavailable, never silently set to zero. Shared interpolation endpoints do not create independent information.

The engineering predictions define future falsifiable comparisons, not validation evidence. No named implementation, data product, observing run, or executable path was assessed in this targeted revision.

Compact rationale-to-contract crosswalk

Narrative topic	Engineering authority
Identity, checked time, grid	EQ-001–003; REQ-002–015; PRED-001, 002, 009
Paired x/r relation	EQ-006–007; REQ-001, 004–006; PRED-019–021
Field operators and telescope families	EQ-004–005; REQ-016–023; PRED-005, 006, 022, 025
Gaps, origin, exposure, intervals	EQ-008–010, 018–020; REQ-024–036; PRED-007, 008, 010, 014, 026
Transfer, response, covariance	EQ-011–017; REQ-037–055; PRED-004, 012, 013, 023, 024

Owner decision register summary

Open or deferred decisions remain: producer event/epoch and offset domains (ODQ-101); exact observing-state registry (102); physical scan/**Hold** semantics (103); detector continuity-surrogate policy (104); response, covariance, and detail tiers (105); exact HWPR registry and all polarization science (109); and exact pointing-correction registry (110). Decisions 106–108 remain closed as recorded. Each unresolved item blocks only its dependent field, operation, or claim.

Final status. This targeted draft records scientific semantics for owner review. Implementation conformity, validation, scientific approval, freeze, readiness, and production authorization remain unassessed.